

Plasminogen Is Tethered with High Affinity to the Cell Surface by the Plasma Protein, Histidine-rich Glycoprotein*

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Plasminogen has been implicated in extracellular matrix degradation by invading cells, but few high affinity cell surface receptors for the molecule have been identified. Previous studies have reported that the plasma protein, histidine-rich glycoprotein (HRG), interacts with plasminogen and cell surfaces, raising the possibility that HRG may immobilize plasminogen/plasmin to cell surfaces. Here we show, based on optical biosensor analyses, that immobilized HRG interacts with soluble plasminogen with high affinity and with an extremely slow dissociation rate. Furthermore, the HRG-plasminogen interaction is lysine-dissociable and involves predominantly the amino-terminal domain of HRG, and the fifth kringle domain of plasminogen, but not the carboxyl-terminal lysine of HRG. HRG was also shown to tether plasminogen to cell surfaces, with this interaction being potentiated by elevated Zn^{2+} levels and low pH, conditions that prevail at sites of tissue injury, tumor growth, and angiogenesis. Based on these data we propose that HRG acts as a soluble adaptor molecule that binds to cells at sites of tissue injury, tumor growth, and angiogenesis, providing a high affinity receptor for tethering plasminogen to the cell surface and thereby enhancing the migratory potential of cells.

Plasminogen is a plasma protein, which, when converted to plasmin, is known to play a pivotal role in fibrinolysis (1, 2). Plasminogen is a 92-kDa modular glycoprotein consisting of a preactivation peptide, five kringle domains, and a catalytic carboxyl-terminal serine proteinase domain (3), with healthy adults having a plasma concentration of 150–200 $\mu\text{g/ml}$ (~ 1.5 – $2 \mu\text{M}$) (4). Plasmin, the active form of plasminogen, is the proteolytic enzyme responsible for degrading fibrin clots and plays an important role in maintaining vascular homeostasis (5). Earlier studies also indicate that the plasminogen system aids directional cell migration associated with embryogenesis, development, tissue remodeling (6–8), inflammation, angiogenesis, and tumor metastasis (9). This role in cell migration depends on the capacity of plasmin either to de-

grade a number of extracellular matrix proteins directly or to activate other proteases with matrix-degrading capabilities (10–14). Efficient cell migration and/or invasion usually requires that degradative enzymes be expressed on the surface of cells rather than being secreted into the extracellular environment. However, few high affinity plasminogen receptors have been identified on cells, which raises the question of how plasminogen is tethered onto the cell surface. A number of low affinity ($K_d \sim 1 \mu\text{M}$) receptors for plasminogen have been defined, such as α -enolase (15), gangliosides (16), and annexin-II (17). Indeed, results arising from *in vivo* studies in annexin-II-deficient mice demonstrated that annexin-II is an important regulator of cell surface plasmin generation, particularly in the absence of fibrin (18). Furthermore, it has been argued that because plasminogen is present in plasma at $\sim 2 \mu\text{M}$, it could theoretically associate with cells via these low affinity receptors. It is known, however, that plasmin is inactivated rapidly unless it remains bound to cells (1), and thus a low affinity interaction with high rates of dissociation would result in plasminogen being inactivated rapidly. This would suggest that high affinity plasminogen receptors are essential if plasmin is to aid cell migration effectively.

Histidine-rich glycoprotein (HRG)¹ is a 75-kDa plasma protein of relatively high abundance (100–200 $\mu\text{g/ml}$, $\sim 2 \mu\text{M}$) characterized by its high content of histidine and proline residues, each accounting for $\sim 13\%$ of total amino acids (19). It is a multidomain molecule consisting of two cystatin-like domains at the amino terminus (termed N1 and N2), a central histidine-rich region (HRR) flanked by two proline-rich sequences, and a carboxyl-terminal domain. Interestingly, HRG is reported to bind a variety of ligands, including plasminogen (20, 21), Zn^{2+} and heme (22–24), fibrinogen (25), thrombospondin (26), IgG (27, 28), C1q (28, 29), and heparin (20, 30–32), and can interact with various cell surface receptors (33–35). Significantly, HRG is known to interact with both plasminogen and cell surfaces, raising the possibility that HRG may immobilize plasminogen/plasmin to the cell surface.

In the present study we examined the interaction between HRG and plasminogen using both optical biosensor and cell surface binding assays. Our findings provide the first evidence that HRG can tether plasminogen to the surface of cells, specifically at low pH and in the presence of $20 \mu\text{M}$ Zn^{2+} , conditions

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¹ The abbreviations used are: HRG, histidine-rich glycoprotein; ABTS, 2,2'-azino-bis(3-ethylbenzthiazolinesulfonic acid); BSA, bovine serum albumin; EDC, 1-ethyl-3-(3-dimethylaminopropyl)carbodiimide; ELISA, enzyme-linked immunosorbent assay; His₆, hexahistidine; HRR, histidine-rich region of HRG; LBS, ligand binding sites; mAb, monoclonal antibody; Ni-NTA, nickel-nitrilotriacetic acid; N1N2, the amino-terminal domain of HRG; PBS, phosphate-buffered saline.

that usually occur at sites of tissue injury, inflammation, tumor growth, and angiogenesis. Based on these findings we propose that under conditions of low pH and elevated free Zn^{2+} , HRG binds to cell surfaces and acts as a high affinity receptor for plasminogen. This would, in theory, enhance the efficiency of conversion of plasminogen to plasmin, protect plasmin from inactivation, and provides a mechanism for polarizing the proteolytic activity of plasmin on the cell surface, resulting in enhanced migratory potential.

EXPERIMENTAL PROCEDURES

Cell Lines—B16F1 cells were cultured in RPMI 1640 medium (Invitrogen) supplemented with 10% fetal calf serum and incubated at 37 °C in a humidified atmosphere containing 5% CO_2 . The *Spodoptera frugiperda*-derived insect cell line Sf9 was cultured in Sf-900 II serum-free medium (Invitrogen) at 27 °C.

Purification of Human HRG, Plasminogen, and Angiostatin—Native human HRG was purified from fresh human plasma as described previously (36). Briefly, a phosphocellulose column was prepared and equilibrated with loading buffer (0.5 M NaCl, 10 mM sodium phosphate, 1 mM EDTA, pH 6.8) for 24 h. Fresh human plasma was provided by Red Cross House, The Canberra Hospital, Canberra, Australia, and mixed with the protease inhibitors aprotinin (2 μ g/ml), phenylmethylsulfonyl fluoride (100 μ g/ml) and 4-(2-aminoethyl)-benzenesulfonyl fluoride hydrochloride (100 μ g/ml). The plasma was passed through the equilibrated column, unbound protein was removed by extensive washing with loading buffer, and bound HRG was eluted with 2.0 M NaCl, 10 mM sodium phosphate, 1 mM EDTA, pH 6.8. Native human plasminogen was purified from fresh human plasma, as described previously (37–39), by passage of plasma through a lysine-Sepharose 4B column (Amersham Biosciences). The column was equilibrated with 0.1 M sodium phosphate, pH 7.4, and was washed extensively with 0.3 M sodium phosphate, pH 7.4, before elution of bound plasminogen with 0.1 M ϵ -aminocaproic acid, pH 7.4. Purified proteins were stored at –70 °C. Angiostatin was purified as described previously (40).

Plasmid Constructs—Using the pUC9-HRG plasmid construct as a template (19), PCR with oligonucleotides ALJ-1 and ALJ-5 was used to amplify a cDNA encoding the full-length human HRG comprised of the leader, first cystatin (N1) domain, second cystatin (N2) domain, the HRR, and carboxyl-terminal domain coding regions. A mutant form of HRG (C-mutant HRG) cDNA consisting of the full-length HRG coding region lacking the carboxyl-terminal lysine codon was generated by amplification with oligonucleotides ALJ-1 and ALJ-4. The N1 coding region was generated by amplification with oligonucleotides ALJ-1 and ALJ-2, and a cDNA encoding the first two cystatin domains (N1N2) was generated by amplification with oligonucleotides ALJ-1 and ALJ-3. Both the N1 and N1N2 cDNAs were engineered to encode a His₆ tag at their 3'-end through inclusion of 6 tandem histidine codons in oligonucleotides ALJ-2 and ALJ-3. The oligonucleotide primer sequences used were as follows: ALJ-1, 5'-TTGAATTCTAAAGGGATGGTTTAAACAAATG-3'; ALJ-2, 5'-AAGGTACCTTAGTGATGGTGATGGTGATGAGAACTTGGTGCAGTTATTAAG-3'; ALJ-3, 5'-AAGGTACCTTAGTGATGGTGATGGTGATGGTGAATGCGAAGGTCAC-3'; ALJ-4, 5'-AAGGTACCTTATGGAATGTATGTGTAACAAACATG-3'; ALJ-5, 5'-AAGGTACCTTATTTGGAAATGTATGTGTAACAAAC-3'. The amplified HRG and HRG fragment cDNAs were cloned into the EcoRI/KpnI sites in the donor vector construct pFastBac (Invitrogen), generating pFastBac-HRG, pFastBac-C-mutHRG, pFastBac-N1, and pFastBac-N1N2 expression constructs, respectively. The nucleotide integrity of each clone was confirmed by automated sequencing using a 3730 DNA Analyzer (Applied Biosystems).

Transfections and Recombinant Protein Production Using the Baculovirus Expression System—Recombinant proteins of full-length HRG, C-mutant HRG, the N1 fragment, and the N1N2 domain of human HRG were produced using the "Bac-to-Bac" baculovirus expression system (Invitrogen). Recombinant bacmid constructs were generated using the pFastBac-HRG, pFastBac-C-mutHRG, pFastBac-N1, or pFastBac-N1N2 constructs according to the manufacturer's instructions. Briefly, Sf9 cells were transfected using Cellfectin reagent (Invitrogen), with 1–2 μ g of recombinant bacmid DNA being transfected/9 $\times 10^5$ cells for 5 h at 27 °C. Supernatant containing recombinant baculovirus was harvested 72 h post-transfection and was amplified for 3–4 days by infecting Sf9 cells with a multiplicity of infection of ~0.01–0.1. Typically, recombinant baculovirus was amplified three times before being used to produce recombinant protein. Recombinant protein was purified from harvested Sf9 cell supernatants by passage of the supernatants

through a Ni-NTA-agarose column (Qiagen). Recombinant full-length HRG, C-mutant HRG, and recombinant His₆-tagged N1 and N1N2 were then eluted from the Ni-NTA-agarose with 200 mM imidazole. Proteins were stored at –70 °C until use.

Western Blotting—Recombinant proteins were boiled for 10 min in 20 μ l of SDS reducing sample buffer (125 mM Tris-HCl, pH 6.8, 20% glycerol, 10% dithiothreitol, 4% SDS) and then subjected to SDS-PAGE analysis on a 4–20% gradient (w/v) precast polyacrylamide MiniGel (Gradipore, Sydney, Australia). Proteins were transferred electrophoretically using a Mini-Protein II apparatus (Bio-Rad) onto a nitrocellulose membrane using a transfer buffer containing 48 mM Tris, 39 mM glycine in 20% (v/v) methanol. The membrane was blocked overnight with 5% (w/v) skim milk powder in PBS. Full-length HRG, C-mutant HRG, and the N1N2 domain were detected using a HRG-specific (HRG-4; AGEN, Brisbane, Australia) mAb, and N1 and N1N2 were detected using a His₆-specific (Roche Diagnostics) mAb (His₆). Chemiluminescence was detected using ECL Western blotting detection reagents (Amersham Biosciences).

Enzyme-linked Immunosorbent Assays (ELISAs)—ELISAs were performed by coating 96-well PVC microtiter plastic plates (Dynax Technologies Inc., Chantilly, VA) overnight at 4 °C with plasminogen or recombinant full-length HRG, C-mutant HRG, N1 fragment, or N1N2 domain (50 μ l/well, ~5 μ g/ml) diluted in a 0.05 M Na_2CO_3 /NaHCO₃ buffer, pH 9.6 (Sigma). Plates were then washed in PBS and 0.01% Tween 20 (Sigma) and blocked for 120 min at room temperature with 3% (w/v) BSA diluted in PBS. For the plasminogen-pretreated plates, recombinant forms of HRG diluted in PBS and 1% BSA (50 μ l/well, ~5 μ g/ml) were added to the plates, allowed to bind for 60 min at room temperature, and then washed in PBS and 0.01% Tween 20. HRG and HRG fragment binding to empty blocked plates was used as the negative background control. Bound full-length HRG, C-mutant HRG, and the N1N2 domain were detected using HRG-4, whereas bound N1 fragment was detected using His₆, followed by secondary antibody detection with a sheep anti-mouse Ig horseradish peroxidase conjugate (Amrad Biotech, Melbourne, Australia). Plate-bound peroxidase was detected using ABTS peroxidase substrate (Kirkegaard and Perry Laboratories Inc. Gaithersburg, MD) by measuring absorbance at 405 nm (reference wavelength 490 nm) on a Thermomax microplate reader. The data were analyzed using SoftMaxPro software (Molecular Devices Corp., Sunnyvale, CA).

Immunofluorescence Flow Cytometry—Cell lines were analyzed for HRG or plasminogen/angiostatin cell surface binding by flow cytometry. Typically, 100 μ g/ml plasma-derived or recombinant HRG and/or 150 μ g/ml plasma-derived plasminogen/angiostatin was added to 5 $\times 10^5$ cells in PBS and 0.1% BSA, pH 7.2, $\pm 20 \mu$ M Zn^{2+} for 60 min at 4 °C and washed three times with PBS and 0.1% BSA. Zinc acetate was used as a source of Zn^{2+} , and Zn^{2+} was added after BSA to PBS to prevent precipitation. Cell-bound HRG was detected using the HRG-4 mAb followed by secondary detection with sheep anti-mouse Ig fluorescein isothiocyanate (Amrad Biotech), and cell-bound plasminogen or angiostatin was detected using a rabbit polyclonal plasminogen Ab (DAKO A/S, Glostrup, Denmark) followed by secondary detection with sheep anti-rabbit Ig fluorescein isothiocyanate (Amrad Biotech). Samples incubated with the relevant primary and secondary detection antibodies and/or conjugates in the absence of HRG or plasminogen/angiostatin served as a background binding control. Cells were analyzed by immunofluorescence flow cytometry using a LSR Flow Cytometer (BD Biosciences), and the resultant data were analyzed using Cell Quest Pro software (BD Biosciences). Each condition was typically performed in triplicate, and each experiment was repeated two or three times. In some experiments, cells were incubated with HRG in PBS and 0.1% BSA with different Zn^{2+} concentrations in the range 0–100 μ M, and/or pH in the range pH 6.0–8.0.

Surface Plasmon Resonance—A BIAcore 2000 (Pharmacia Biosensor) instrument was used to measure binding by surface plasmon resonance. CM5 sensor chips and coupling reagents were purchased from Pharmacia Biosensor. The carboxyl methylated dextran surfaces of flow-cells on the CM5 chip were activated by injecting 35 μ l of 0.05 M EDC and 0.05 M *N*-hydroxysuccinimide diluted in double distilled Milli-Q water at 5 μ l/min. Plasma-derived and recombinant HRG each were immobilized onto the *N*-hydroxysuccinimide ester-activated surface by injecting ~10 μ l of HRG (100 μ g/ml) diluted in 10 mM sodium acetate, pH 4.5, at 5 μ l/min. Remaining *N*-hydroxysuccinimide esters were deactivated by injecting 35 μ l of 1 M ethanolamine, pH 8.5, at 5 μ l/min. Typically, a flow rate of 10 μ l/min was used throughout the binding assays. Different concentrations of plasminogen/angiostatin (0–2,000 nM, 30–100 μ l) were injected into the flow-cells of the biosensor, with binding and dissociation each monitored for 3–10 min. Plasminogen-HRG complexes

were effectively disrupted by injecting 100 mM L-lysine for 1 min at a flow rate of 100 μ l/min. Experiments used an automated program to control triplicate injections of each binding protein. Flow-cell 1 was used as a blank reference cell, with the background binding to the dextran matrix detected in flow-cell 1 being subtracted from responses in flow-cells 2, 3, and 4. Binding curves were analyzed using the BIA Evaluation program (Pharmacia Biosensor).

RESULTS

Effect of Zn^{2+} and pH on the Binding of Human HRG to Cell Surfaces—Previous studies have shown that the binding of HRG to surface-immobilized heparin is dependent on both Zn^{2+} and pH, with Borza and Morgan (41) suggesting that Zn^{2+} and pH act synergistically to affect the conformation and thus function of HRG. This observation prompted us to test the effect of Zn^{2+} and pH on HRG binding to cell surfaces. We used the metastatic mouse melanoma cell line, B16F1, in the cell surface binding studies. Dose-response experiments indicated that the potentiating effect of Zn^{2+} on HRG binding to B16F1 cells plateaued between 5 and 50 μ M but was near maximal at ~ 20 μ M Zn^{2+} (Fig. 1, A and B), with HRG binding being enhanced ~ 6 -fold. Such concentrations of Zn^{2+} can occur physiologically (42). In contrast, at high Zn^{2+} concentrations (100 μ M), HRG binding was decreased to less than that observed in the absence of any added Zn^{2+} (Fig. 1B). A similar effect of Zn^{2+} concentration on HRG binding was observed with CHO-K1 cells (data not shown). We also investigated the effect of pH on HRG binding to cell surfaces in the presence of an optimal Zn^{2+} concentration (20 μ M) and found that binding was minimal at pH 8.0 but increased steadily as the pH was reduced to 6.0 (Fig. 1, C and D). In fact, there was a ~ 250 -fold increase in the level of cell surface-bound HRG across this pH range. Thus, it appears that both Zn^{2+} and pH directly affect the interaction of HRG with cell surfaces, with optimal HRG binding occurring in the presence of ~ 20 μ M Zn^{2+} and at pH 6.0.

Soluble Plasminogen Binds to Immobilized HRG—Previous studies identified plasminogen as an important HRG ligand (20, 21). Using the optical biosensor, we further characterized this interaction by monitoring the binding and dissociation of soluble plasminogen to immobilized HRG (Fig. 2A). These binding studies showed that the interaction of soluble plasminogen with immobilized HRG is dependent on the plasminogen concentration over the range tested of 32–2,000 nM (3–185 μ g/ml). The almost flat dissociation curve (Fig. 2A) indicates an extremely slow off-rate or dissociation between HRG and plasminogen, reflecting a high affinity interaction (see analysis below). In fact, even when dissociation was allowed to occur for many hours, plasminogen remained tightly bound to HRG (data not shown). Nevertheless, bound plasminogen could be very effectively displaced by exposure to 100 mM L-lysine, indicating that lysine can disrupt HRG-plasminogen complexes, and hence the interaction is likely to be mediated by lysine residues (Fig. 2B). Alternatively, lysine may disrupt the plasminogen-HRG complex by binding to plasminogen and inducing a conformation change that results in the complex dissociating. Similar association-dissociation results were obtained with plasminogen-I and plasminogen-II (data not shown), isoforms of the molecule that differ by one glycosylation site (43–45).

An analysis of the binding curves for the interaction between soluble plasminogen and immobilized HRG using the BIA-Evaluation program showed that the data could be described by both a bivalent analyte model and a two-state conformation model. The bivalent analyte model assumes that each molecule of the analyte (plasminogen) can interact bivalently with one or two immobilized ligand (HRG) molecule(s). Binding to the first HRG molecule is defined by a single set of rate constants (k_{a1}

and k_{d1}), and binding to the second HRG molecule is defined by a second set of rate constants (k_{a2} and k_{d2}), thus allowing the model to take cooperative effects into account (Table I). Such cooperative effects could well explain the extremely slow off-rates that are observed (Fig. 2). The two-state conformation model, on the other hand, assumes that one molecule of plasminogen binds to one or more molecules of immobilized HRG, and that this is then followed by a conformation change (represented as * in the reactions below) in the complex, which stabilizes the interaction. Again, two sets of kinetic constants are produced (Table I), involving two-step association and dissociation, shown below.



REACTION 1



REACTION 2



REACTION 3



REACTION 4

Both the bivalent analyte and two-state conformation models estimate an apparent K_d value for the interaction of ~ 200 –300 nM, with the most notable feature of the interaction between immobilized HRG and soluble plasminogen being the extremely slow dissociation/off-rate, indicating a high affinity, highly stable interaction.

Plasminogen Binding to the Surface of B16F1 Cells Is Enhanced Dramatically in the Presence of HRG—Although plasminogen has been reported to interact with HRG, there has been no previous indication that HRG can potentiate the interaction of plasminogen with the cell surface. Using flow cytometry, we found that 150 μ g/ml plasminogen binding to the surface of B16F1 cells is enhanced greatly (up to 14-fold) when the cells have been either precoated with 100 μ g/ml plasma-derived HRG (Fig. 3A, HRG/plasminogen) or when plasminogen is coincubated with HRG (Fig. 3A, HRG+plasminogen). The effect of HRG on plasminogen binding to B16F1 cells was maximal when the cells were coincubated with a mixture of plasminogen and HRG, rather than precoating the cells with HRG prior to plasminogen exposure ($p < 0.01$, Fig. 3B). HRG also enhanced the binding of plasminogen to cells over a wide range of plasminogen concentrations (0.1–200 μ g/ml) (data not shown), although these studies showed that in the presence of physiological concentrations of HRG (100 μ g/ml, 1.3 μ M), maximal plasminogen binding occurred with 150 μ g/ml (~ 1.6 μ M) plasminogen, which is the physiological plasma concentration of this molecule. Further studies aimed at characterizing the effect of soluble HRG on plasminogen cell surface binding showed that excess soluble HRG (>5 times molar excess of HRG over plasminogen) (1.6 μ M plasminogen, 8 μ M HRG) did not inhibit cell surface plasminogen binding (Fig. 3C). Even relatively low concentrations of HRG (0.3 μ M) were sufficient to support near maximum plasminogen binding, suggesting that HRG is an efficient adaptor molecule for binding plasminogen to the cell surface. Our initial studies indicated that pH affects HRG cell surface binding (Fig. 1, C and D). Using flow cytometry, we found that the ability of HRG to potentiate the binding of plasminogen to cells was pH-dependent (Fig. 3D). Thus, plasminogen binding steadily increased with decreasing pH, the lowest plasminogen binding occurring at pH 8.0, and the highest at pH 6.0. At all pH values tested,

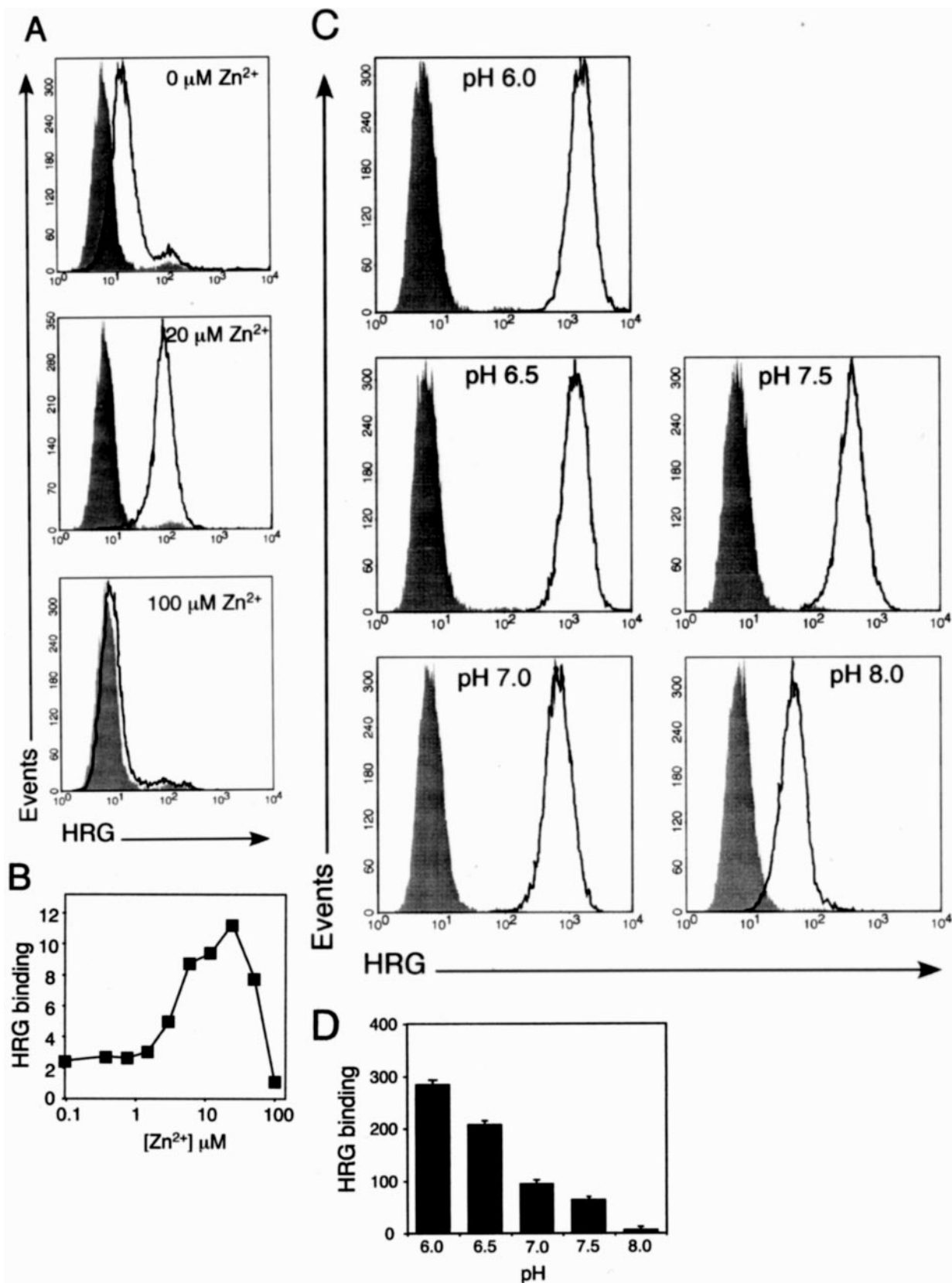


FIG. 1. Effect of Zn^{2+} and pH on HRG binding to B16F1 melanoma cells. A, B16F1 cells were incubated with 100 $\mu\text{g}/\text{ml}$ plasma-derived HRG in PBS and 0.1% BSA, pH 7.2, in the presence of 0, 20, or 100 μM Zn^{2+} and then analyzed for HRG binding using the HRG-specific mAb HRG-4 and flow cytometry. Representative flow cytometry histograms show HRG binding at 0, 20, and 100 μM Zn^{2+} . Shaded histograms represent background binding of HRG-4 in the absence of HRG; open histograms represent HRG binding. B, numerical values showing HRG binding to B16F1 cells as a fold increase in median fluorescence above background at different Zn^{2+} concentrations (0–100 μM). C, B16F1 cells were incubated with 100 $\mu\text{g}/\text{ml}$ plasma-derived HRG in PBS, 0.1% BSA, and 20 μM Zn^{2+} at various pH values (6.0–8.0) and then analyzed for HRG binding by flow cytometry. Representative flow cytometry histograms show HRG binding at pH 6.0, 6.5, 7.0, 7.5, and 8.0. Shaded histograms represent background binding of HRG-4 in the absence of HRG; open histograms represent HRG binding. D, numerical values showing HRG binding to B16F1 cells between pH 6.0 and 8.0 as a fold increase in median fluorescence above background. Data are the mean $\pm$ S.E. of three determinations.

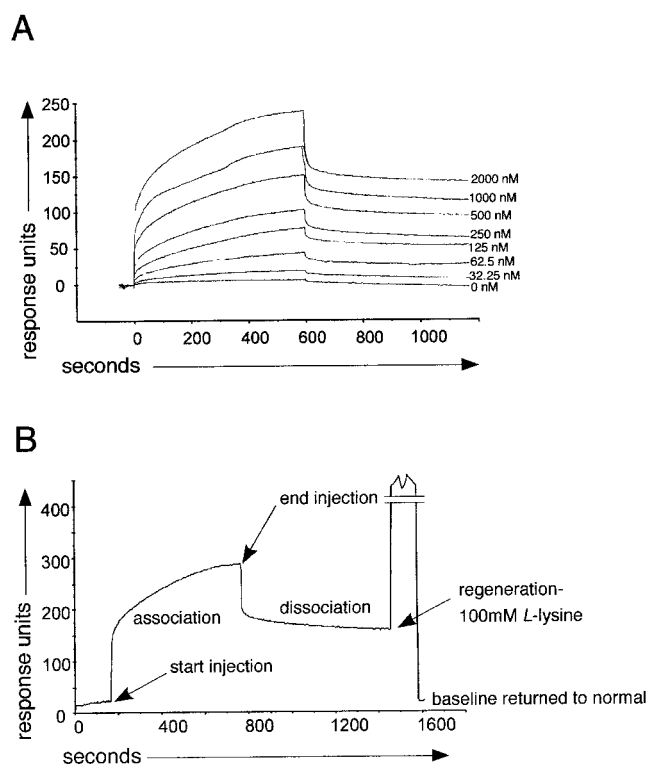


FIG. 2. Binding of soluble plasminogen to immobilized HRG. A, the binding of human plasminogen to human plasma-derived HRG was examined using a BIAcore 2000 biosensor. Plasma-derived human HRG was covalently attached to the carboxyl methylated dextran surface of a CM5 sensor chip, and human plasminogen at the indicated concentrations (diluted in PBS, pH 7.2) was injected into the flow-cells, with binding being monitored for 10 min, followed by a 10-min dissociation period. A flow-cell within the CM5 sensor chip with no immobilized HRG served as a control. The figure shows representative sensorgrams for each plasminogen concentration used. B, displacement of plasminogen bound HRG by 100 mM L-lysine, this treatment being routinely used to regenerate the sensor chip after assessment of HRG binding and dissociation.

plasminogen binding was virtually totally dependent on the presence of HRG (Fig. 3D). These data suggest that pH modulates plasminogen binding by regulating the interaction of HRG with the cell surface.

Analysis of the Regions of HRG and Plasminogen Involved in the HRG-Plasminogen Interaction—Lysine residues have been predicted to play an important role in the interaction between HRG and plasminogen, the biosensor studies outlined in Fig. 2 supporting this view (46). It has been suggested that the carboxyl-terminal lysine residue on HRG is likely to be a candidate as one of the key plasminogen-binding residues (46). On the other hand, it has been proposed that the amino-terminal region of HRG may interact with plasminogen (46). To test these possibilities, we produced in insect cells using a baculovirus expression system, recombinant full-length HRG, C-mutant HRG lacking the carboxyl-terminal lysine residue, the amino-terminal fragment of HRG, termed N1N2, and the first cystatin domain of HRG, termed N1 (Fig. 4A). A His₆ tag was engineered onto the carboxyl terminus of N1 and N1N2 to allow purification of the recombinant proteins by Ni-NTA chelation chromatography. The recombinant full-length HRG and C-mutant HRG were made untagged because they contain sufficient histidine residues in their HRR to allow Ni-NTA-agarose affinity purification. Western blot analysis of purified recombinant full-length HRG (75 kDa), C-mutant HRG (75 kDa), and the N1N2 domain (35 kDa) showed a single band at the appropriate molecular mass using the

HRG-specific mAb, HRG-4, whereas N1 (18 kDa) and N1N2 (35 kDa) also showed dominant bands at the anticipated molecular mass using the His₆ tag-specific mAb, His₆ (Fig. 4B). N1 was not detected by HRG-4 (data not shown). An Sf9 cell culture supernatant was purified using Ni-NTA-agarose and was included as a negative control. ELISA studies (Fig. 4C) indicated that full-length HRG, C-mutant HRG, and the N1N2 domain were recognized by HRG-4, whereas N1 was only detected by His₆, suggesting that the epitope recognized by the HRG-4 mAb is located within the N2, and not the N1 portion of the N1N2 domain. Furthermore, these studies indicate that the HRG-4 mAb interacts with both denatured (Western blotting studies) and native (ELISA data) forms of HRG, C-mutant HRG, and N1N2.

The recombinant full-length HRG and C-mutant HRG were then tested for their ability to bind plasminogen. The two forms of HRG were immobilized onto the surface of a CM5 sensor chip, and the binding of soluble plasminogen was determined by the biosensor. Resultant biosensor sensorgrams (Fig. 5A) showed that the binding of plasminogen to full-length HRG is essentially identical to the binding of plasminogen to C-mutant HRG. Because of high background binding of soluble HRG we were, unfortunately, unable to use the biosensor to examine the binding of soluble HRG to immobilized plasminogen. As an alternative approach, we immobilized plasminogen in the wells of plastic microtiter plates and used an ELISA to measure the binding of both full-length HRG and C-mutant HRG to the immobilized plasminogen. Again, these experiments indicated that plasminogen binds to C-mutant and full-length HRG with similar affinity, and thus the carboxyl-terminal lysine residue on HRG does not appear to be essential for plasminogen binding (Fig. 5B). To investigate further the plasminogen binding domain within HRG, we also tested whether the recombinant N1 fragment or the N1N2 domain could bind to immobilized plasminogen. ELISA studies indicated that the N1N2 domain exhibited significant binding to plasminogen ($p < 0.01$), although binding appears to be somewhat lower than full-length recombinant HRG, whereas the smaller fragment, N1, failed to bind to plasminogen ($p = \text{not significant}$) (Fig. 5C). Biosensor studies with immobilized N1N2 or N1 and soluble plasminogen were unsuccessful because of the poor immobilization of these HRG fragments to the surface of the biosensor chips.

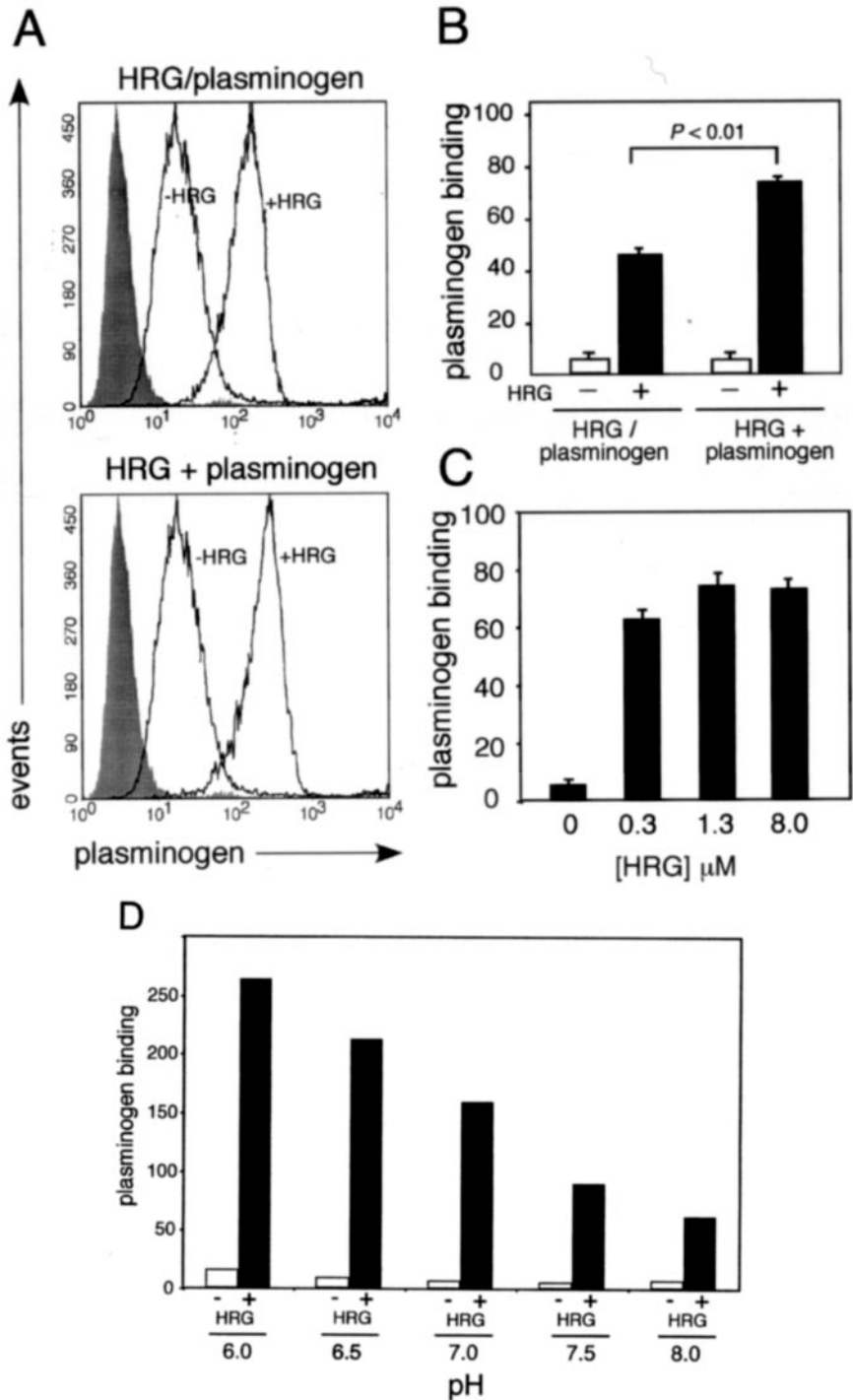
Angiostatin, which consists of the first four kringle domains and part of the fifth kringle domain of plasminogen, is a fragment of plasminogen which exhibits antiangiogenic activity. Because the present study has shown that plasminogen binds with high affinity to HRG, it was of interest to determine whether this truncated form of plasminogen also exhibited HRG binding. Based on optical biosensor studies, it was found that angiostatin did not bind with high affinity to immobilized HRG compared with plasminogen, 2 μM angiostatin exhibiting rapid association and dissociation, a sensorgram typical of a low affinity interaction. In fact, a component of the intact plasminogen sensorgram involves rapid binding and dissociation (Fig. 6A, upper panel), this aspect of the plasminogen-HRG interaction being retained by angiostatin, *i.e.* very rapid association and dissociation rates (Fig. 6A). Similarly, immunofluorescence flow cytometry studies indicated that the presence of physiological concentrations of HRG (100 $\mu\text{g}/\text{ml}$) could dramatically enhance the binding of plasminogen to the surface of B16F1 melanoma cells, whereas HRG did not promote the binding of angiostatin to the surface of B16F1 cells (Fig. 6B). It should be noted that ELISA studies, using immobilized angiostatin, demonstrated that the polyclonal anti-plasminogen antibody used in the cell binding studies reacted strongly with angiostatin (data not shown).

TABLE I
Association rate (k_a) and dissociation rate constants (k_d) for the bivalent analyte model and the two-state conformation model for describing the interaction of soluble plasminogen with immobilized HRG

Model	k_{a1}	k_{d1}	k_{a2}^a	k_{d2}
Bivalent	$4.6 \times 10^4 \pm 1.42 \times 10^3 \text{ ms}^{-1}$	$0.19 \pm 0.018 \text{ s}^{-1}$	$1.8 \times 10^{-5} \pm 1.8 \times 10^{-6} \text{ RU}^{-1}$	$1.2 \times 10^{-3} \pm 6.3 \times 10^{-5} \text{ s}^{-1}$
Two-state	$3.1 \times 10^5 \pm 1.8 \times 10^5 \text{ ms}^{-1}$	$0.19 \pm 0.057 \text{ s}^{-1}$	$5.2 \times 10^{-3} \pm 2.9 \times 10^{-4} \text{ s}^{-1}$	$2.5 \times 10^{-3} \pm 1.3 \times 10^{-4} \text{ s}^{-1}$

^a RU, response units.

FIG. 3. Plasminogen binding to the surface of cells is enhanced dramatically in the presence of HRG. A, B16F1 cells were preincubated with or without 100 $\mu\text{g/ml}$ plasma-derived HRG in PBS and 0.1% BSA (pH 7.2, 20 μM Zn^{2+}), washed with PBS and 0.1% BSA, incubated with 150 $\mu\text{g/ml}$ plasminogen in PBS and 0.1% BSA, pH 7.2, and then analyzed for plasminogen binding by flow cytometry (upper panel, HRG/plasminogen). Alternatively, B16F1 cells were incubated simultaneously with 100 $\mu\text{g/ml}$ plasma-derived HRG and 150 $\mu\text{g/ml}$ plasminogen in PBS, 0.1% BSA (pH 7.2, 20 μM Zn^{2+}) and then analyzed for plasminogen binding by flow cytometry (lower panel, HRG + plasminogen). Representative flow cytometry histograms show plasminogen binding $\pm$ HRG, with shaded histograms representing background binding of the polyclonal plasminogen antibody in the absence of plasminogen; open histograms represent plasminogen binding. B, quantitative comparison, based on data from A, of the effect of preincubating cells with HRG (HRG/plasminogen) versus the simultaneous incubation of cells with HRG and plasminogen (HRG + plasminogen) on plasminogen binding to cells. Plasminogen binding is expressed as a -fold increase in median fluorescence relative to background fluorescence, with error bars being S.E. ($n = 3$). For each treatment condition, white histograms represent plasminogen binding in the absence of HRG, and black histograms represent plasminogen binding in the presence of HRG. C, B16F1 cells were incubated with 150 $\mu\text{g/ml}$ plasminogen (1.6 μM) and 0, 0.3, 1.3, or 8 μM plasma-derived HRG in PBS and 0.1% BSA (pH 7.2, 20 μM Zn^{2+}) and then analyzed for plasminogen binding by flow cytometry. Numerical values show plasminogen binding as a -fold increase in median fluorescence above background binding of plasminogen-specific antibody to cells. D, B16F1 cells were incubated with 150 $\mu\text{g/ml}$ plasminogen and either with (black histogram) or without (white histogram) 100 $\mu\text{g/ml}$ plasma-derived HRG in PBS, 0.1% BSA, 20 μM Zn^{2+} at pH 6.0, 6.5, 7.0, 7.5, and 8.0, and then analyzed for plasminogen binding by flow cytometry. Plasminogen binding is shown as a -fold increase in median fluorescence above background binding of plasminogen-specific antibody to cells.



DISCUSSION

For extracellular matrix-degrading enzymes, such as plasmin, to aid cell invasion optimally they need to be tethered to the surface of invading cells. Many cell types express specific receptors for urokinase plasminogen activator, but to date, few if any high affinity cell surface receptors for plasminogen have

been described, although low affinity ($K_d \sim 1 \mu\text{M}$) receptors including annexin-II (17, 18), α -enolase (15), and gangliosides (16) have been identified. Here we show for the first time that the soluble plasma protein HRG can act as an adaptor protein that tethers plasminogen to cell surfaces in a highly stable manner. Interestingly, the binding of HRG (and thus plasmin-

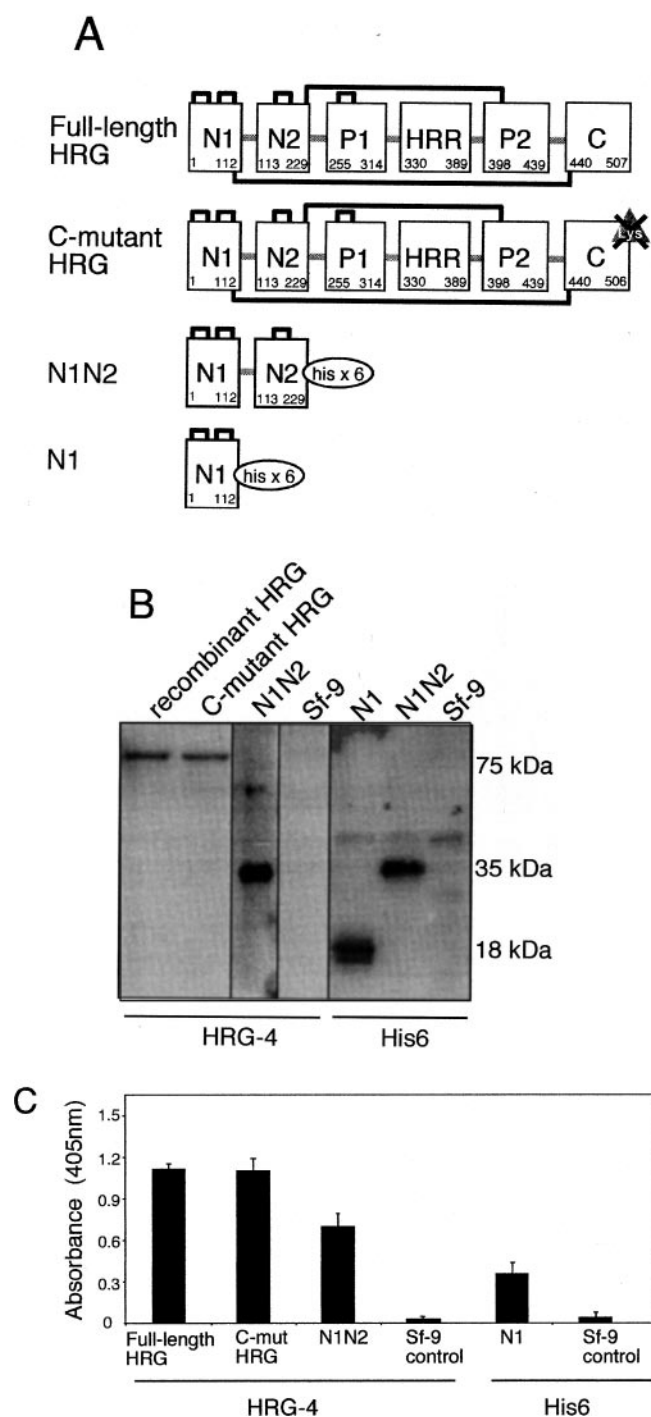


FIG. 4. Production of recombinant forms of HRG. A, schematic representation of the different recombinant forms of HRG, namely full-length HRG; C-mutant HRG, which lacks a terminal lysine residue; the N1N2 amino-terminal domain of HRG; and an N1 fragment, with both N1 and N1N2 containing a carboxyl-terminal His₆ tag to aid purification. Inter- and intradomain disulfide bonds are represented as solid black lines. Recombinant proteins were produced by baculovirus expression in the Sf9 insect cell line. B, Western blot analysis of Ni-NTA-agarose-purified recombinant full-length HRG, C-mutant HRG, and the N1 and N1N2 fragments of HRG on reducing SDS-polyacrylamide gels. Recombinant full-length HRG (75 kDa) and C-mutant HRG (75 kDa) were detected by HRG-4; the N1 fragment (18 kDa) was detected by a mAb specific for its His₆ tag. In contrast, the N1N2 domain (35 kDa) was detected by both the HRG-4 and His₆ mAbs. Ni-NTA-purified Sf9 cell culture supernatant was included as a negative control. C, ELISA showing that recombinant full-length HRG, C-mutant HRG, and the N1N2 domain (~5 µg/ml), when immobilized to plastic in their native state, bind HRG-4, whereas the N1 fragment binds His₆. Ni-NTA-purified Sf9 cell culture supernatant was included as a negative control.

ogen) to the cell surface appears to be dependent on acidic pH and enhanced levels of free Zn²⁺ (Fig. 1). In this regard, Zn²⁺ stored in platelet granules is released rapidly into the plasma after thrombin-stimulated platelet degranulation (47, 48), while pH can drop as much as one pH unit during hypoxia or ischemia, or 0.5 of a pH unit during an inflammatory response because of lactic acidosis. Thus, at sites of inflammation, angiogenesis, and wound healing, acidic pH and elevated free Zn²⁺ provide a mechanism for plasmin activity to be expressed selectively on cell surfaces.

Metal divalent cations, in particular, Zn²⁺, are known to interact with the HRR of HRG (23, 24, 49). The high concentration of histidine residues located within the HRR also results in HRG having an ionic charge that is sensitive to pH in the range between pH 6 and 7 as the histidine residues become protonated (41). Using optical biosensor studies, Borza and Morgan (41) found that HRG binding to immobilized heparin was strikingly pH-sensitive, with maximum binding occurring at pH <6.0. Poor HRG binding was observed at physiological pH in the absence of Zn²⁺, although the interaction was promoted by the addition of free Zn²⁺, and the pH dependence was shifted toward alkaline pH by Zn²⁺ (41). The HRR was suggested to act like a pH sensor, whereby Zn²⁺ and pH act synergistically in regulating HRG function. Consistent with these findings, the present study shows that pH can profoundly alter HRG cell surface binding, whereby HRG binding to B16F1 cells is potentiated greatly at pH 6.0 but reduced at pH 8.0 (Fig. 1, C and D). Of particular relevance here is our recent finding that heparan sulfate is the predominant cell surface ligand for HRG, with Zn²⁺ regulating this interaction and the amino-terminal N1N2 domain of HRG interacting with this ligand (50). Hence, we propose that under conditions of local low pH and free Zn²⁺, the HRR of HRG binds Zn²⁺ and enhances HRG cell surface binding to heparan sulfate via its N1N2 domain. Under such conditions, the modular domain structure of HRG would allow cell surface-bound HRG to coimmobilize other molecules, such as plasminogen, to the cell surface.

Previous studies based on chemical modification of lysines suggest that HRG binds to plasminogen via HRG lysine residues (46). The biosensor studies described herein support the notion that the plasminogen-HRG interaction is dependent on the well described lysine binding sites (LBS) (8) of plasminogen, as free L-lysine reversed the interaction. Because carboxyl-terminal lysine residues of many proteins are often involved in plasminogen binding, recombinant HRG lacking the carboxyl-terminal lysine residue was produced. Surprisingly, our data show that there is no difference in the binding of plasminogen to C-mutant HRG and full-length HRG (Fig. 5, A and B), implying that this residue does not play a role in the interaction. On the other hand, the amino-terminal N1N2 domain of HRG, but not a N1 fragment, interacted with plasminogen (Fig. 5C), although it is possible that lysine residues within other regions of HRG may participate in plasminogen binding.

The heavy chain of plasminogen contains five triple loop structures termed "kringles" which are held in a loop structure by three disulfide bridges (4) and contain one high affinity LBS ($K_d \sim 9 \mu\text{M}$) and four or five low affinity LBS ($K_d \sim 5 \text{ mM}$) that play a crucial role in the regulation of fibrinolysis by interacting specifically with lysine residues on fibrin, α_2 -antiplasmin, and cell surfaces during the physiological lysis of fibrin. Previous studies suggest that HRG binds to the high affinity LBS (20). Analysis of our own optical biosensor data suggests that immobilized HRG interacts bivalently with plasminogen, the interaction being found to have extremely slow off-rates (Fig. 2A), suggesting that cooperative binding and/or a change in

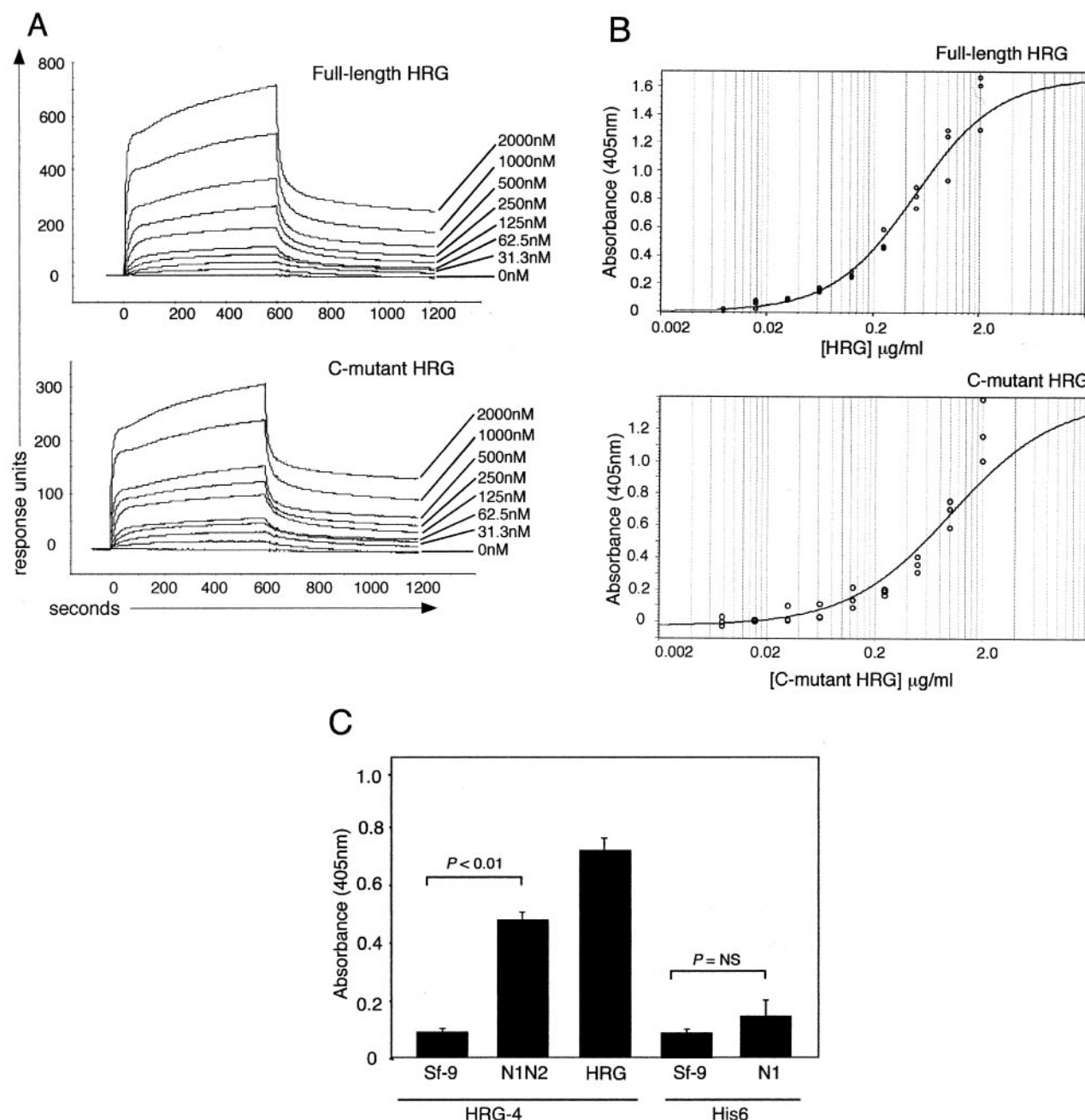


FIG. 5. Analysis of plasminogen binding to recombinant full-length HRG, C-mutant HRG, and the HRG fragments N1 and N1N2. A, recombinant full-length HRG and C-mutant HRG were immobilized on the surface of different flow-cells of a CM5 sensor chip, and human plasminogen, diluted in PBS, pH 7.2, at the indicated concentrations, was then injected into the biosensor flow-cells, with binding and dissociation each being monitored for 10 min. A flow-cell within the chip with no immobilized HRG served as a control. The figure shows representative sensorgrams for each plasminogen concentration. B, plasma-derived plasminogen was immobilized on ELISA plates overnight, and, after blocking of nonspecific binding, various concentrations of recombinant full-length and C-mutant HRG (0.1–2 $\mu\text{g/ml}$) were allowed to bind to the plates. HRG binding was detected with HRG-4, using ABTS detection reagents and with the optical density being measured at 405 nm. C, plasma-derived plasminogen was immobilized on ELISA plates overnight, and after blocking of nonspecific binding, 2 $\mu\text{g/ml}$ recombinant full-length HRG, N1 fragment, or N1N2 domain was allowed to bind to the plates, with binding of full-length HRG or the N1N2 domain being detected using HRG-4 or binding of the N1 fragment being detected using His₆. Antibody binding was detected as in B. Data are the mean $\pm$ S.E. of three determinations.

conformation contributes to the high affinity interaction. It seems highly likely that the two putative HRG binding sites located within plasminogen consist of a high affinity and a low affinity LBS. Our finding that excess soluble HRG does not interfere with the interaction of plasminogen with cell-bound HRG strongly suggests that plasminogen contains at least two HRG binding sites, whereas HRG contains only one site. Thus, a multivalent array of HRG displayed on a cell surface would be able to interact cooperatively with soluble plasminogen to form a highly stable complex. In fact, the demonstration that an-

giostatin binds with only low affinity to HRG (Fig. 6A) suggests that one of the HRG binding sites is LBS-5, because angiostatin lacks a portion of the fifth kringle domain of plasminogen, which contains LBS-5.

In blood, plasminogen circulates in a globular (closed) conformation, probably as a safeguard mechanism to prevent uncontrolled plasmin generation. When bound to a cell or fibrin surface through its LBS, it adopts an extended "open" conformation that is activated more rapidly to form plasmin (3, 4, 51, 52). Borza and Morgan (51) showed previously, using *in vitro*

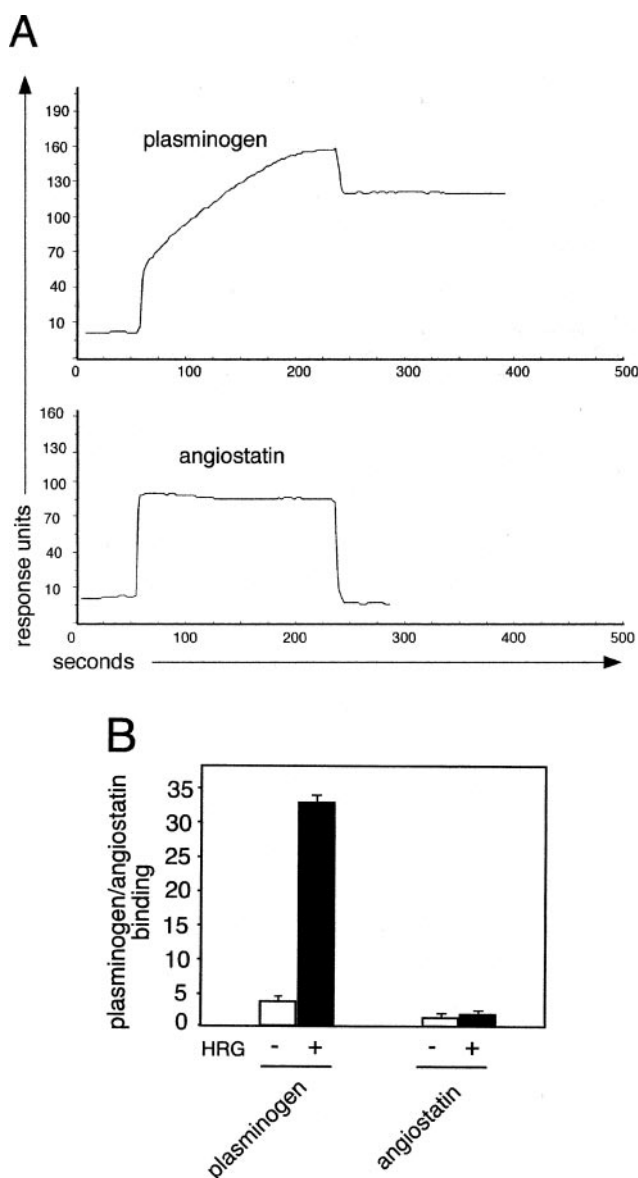


FIG. 6. Angiostatin, a fragment of plasminogen, exhibits low affinity binding to HRG. A, plasma-derived HRG was immobilized onto the surface of a CM5 sensor chip, and either human angiostatin or plasminogen ($2 \mu\text{M}$) diluted in PBS (pH 7.2) was then injected into the biosensor flow-cell, with binding and dissociation each being monitored for 3 min. A flow-cell within the chip with no immobilized HRG served as a control. B, B16F1 cells were incubated simultaneously with angiostatin or plasminogen ($150 \mu\text{g/ml}$) and either with or without $100 \mu\text{g/ml}$ plasma-derived HRG in PBS, 0.1% BSA (pH 7.2, $20 \mu\text{M}$ Zn^{2+}), and then analyzed for angiostatin/plasminogen binding by immunofluorescence flow cytometry. Angiostatin/plasminogen binding is expressed as a -fold increase in median fluorescence relative to background Ab binding, with error bars being the S.E. ($n = 3$). For each treatment condition, white histograms represent plasminogen binding in the absence of HRG, and black histograms represent plasminogen binding in the presence of HRG.

techniques, that plasminogen is activated more rapidly to plasmin when associated with immobilized HRG, but not soluble HRG. Cell-bound plasmin also remains protected from inactivation by α_2 -antiplasmin and α_2 -macroglobulin, whereas soluble plasmin is inhibited rapidly by these proteins (1), ensuring that the generation and activity of plasmin remain localized to the microenvironment of either the clot or the polarized cell surface. Generation of cell surface proteases is fundamental to a wide variety of *in vivo* biological processes, with the plasminogen activator/plasmin system known to be important for pro-

teolysis during cellular migration (10–14, 53). Processes including tumor cell invasion, angiogenesis, embryogenesis, and leukocyte migration to a site of inflammation require cells to invade and penetrate neighboring tissues (17). Directed cell migration requires localized proteolysis, with urokinase plasminogen activator receptors being polarized rapidly to the leading edge of migrating cells to focus plasmin mediated extracellular matrix degradation (13, 14), and indeed, increased urokinase plasminogen activator receptor expression correlates with a poor prognosis for many invasive human cancers (54–56). Through the generation of plasmin, plasminogen activators catalyze the degradation of most proteins of the extracellular space, including laminin, thrombospondin, fibronectin, and fibrinogen (14). Plasmin can also activate other proteases such as matrix metalloproteinases (17). Expression of the powerful degradative potential of plasminogen heavily depends on plasminogen being associated with the cell surface via a high affinity receptor. In this regard it is interesting to note that tumor cells pre-coated with HRG are ~6-fold more metastatic than their untreated counterparts (data not shown), with the cell bound HRG presumably allowing the recruitment of endogenous plasminogen (or other degradative enzymes) to the tumor cell surface. Thus potentially, HRG may provide the means for tumor cells to “hijack” the degradative potential of plasmin and use its proteolytic ability to aid tumor cell metastasis and tumor angiogenesis.

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REFERENCES

- Lijnen, H. R. (2001) *Ann. N. Y. Acad. Sci.* **936**, 226–236
- Collen, D. (1999) *Thromb. Haemostasis* **82**, 259–270
- Parry, M. A., Zhang, X. C., and Bode, I. (2000) *Trends Biochem. Sci.* **25**, 53–59
- Miyashita, C., Wenzel, E., and Heiden, M. (1988) *Haemostasis* **18**, 7–13
- Collen, D. (2001) *Hematology (Am. Soc. Hematol. Educ. Program)* **1**–9
- Dano, K., Andreasen, P. A., Grondahl-Hansen, J., Kristensen, P., Nielsen, L. S., and Skriver, L. (1985) *Adv. Cancer Res.* **44**, 139–266
- Vassalli, J. D., Sappino, A. P., and Belin, D. (1991) *J. Clin. Invest.* **88**, 1067–1072
- Plow, E. F., Herren, T., Redlitz, A., Miles, L. A., and Hoover-Plow, J. L. (1995) *FASEB J.* **9**, 939–945
- Chapman, H. A., Jr., Vavrin, Z., and Hibbs, J. B., Jr. (1982) *Cell* **28**, 653–662
- Chapman, H. A. (1997) *Curr. Opin. Cell Biol.* **9**, 714–724
- Lahteenmaki, K., Kuusela, P., and Korhonen, T. K. (2000) *Methods* **21**, 125–132
- Brunner, G., Reimbold, K., Meissauer, A., Schirmacher, V., and Erkell, L. J. (1998) *Exp. Cell Res.* **239**, 301–310
- Ploplis, V. A., and Castellino, F. J. (2000) *Methods* **21**, 103–110
- Estreicher, A., Muhlhauser, J., Carpentier, J. L., Orci, L., and Vassalli, J. D. (1990) *J. Cell Biol.* **111**, 783–792
- Miles, L. A., Dahlberg, C. M., Plescia, J., Felez, J., Kato, K., and Plow, E. F. (1991) *Biochemistry* **30**, 1682–1691
- Miles, L. A., Dahlberg, C. M., Levin, E. G., and Plow, E. F. (1989) *Biochemistry* **28**, 9337–9343
- Brownstein, C., Falcone, D. J., Jacovina, A., and Hajjar, K. A. (2001) *Ann. N. Y. Acad. Sci.* **947**, 143–156
- Ling, Q., Jacovina, A. T., Deora, A., Febbraio, M., Simantov, R., Silverstein, R. L., Hempstead, B., Mark, W. H., and Hajjar, K. A. (2004) *J. Clin. Invest.* **113**, 38–48
- Koide, T., Foster, D., Yoshitake, S., and Davie, E. W. (1986) *Biochemistry* **25**, 2220–2225
- Lijnen, H. R., Hoylaerts, M., and Collen, D. (1980) *J. Biol. Chem.* **255**, 10214–10222
- Ichinose, A., Mimuro, J., Koide, T., and Aoki, N. (1984) *Thromb. Res.* **33**, 401–407
- Guthans, S. L., and Morgan, W. T. (1982) *Arch. Biochem. Biophys.* **218**, 320–328
- Morgan, W. T. (1981) *Biochemistry* **20**, 1054–1061
- Morgan, W. T. (1985) *Biochemistry* **24**, 1496–1501
- Leung, L. L. (1986) *J. Clin. Invest.* **77**, 1305–1311
- Walz, D. A., Bacon-Baguley, T., Kendra-Franczak, S., and DePoli, P. (1987) *Semin. Thromb. Hemostasis* **13**, 317–325
- Gorgani, N. N., Parish, C. R., and Altin, J. G. (1999) *J. Biol. Chem.* **274**, 29633–29640
- Gorgani, N. N., Parish, C. R., Easterbrook Smith, S. B., and Altin, J. G. (1997) *Biochemistry* **36**, 6653–6662
- Chang, N. S., Leu, R. W., Rummage, J. A., Anderson, J. K., and Mole, J. E. (1992) *Blood* **79**, 2973–2980
- Peterson, C. B., Morgan, W. T., and Blackburn, M. N. (1987) *J. Biol. Chem.*

- 262, 7567–7574
31. Lijnen, H. R., Hoylaerts, M., and Collen, D. (1983) *J. Biol. Chem.* **258**, 3803–3808
32. Burch, M. K., Blackburn, M. N., and Morgan, W. T. (1987) *Biochemistry* **26**, 7477–7482
33. Gorgani, N. N., Smith, B. A., Kono, D. H., and Theofilopoulos, A. N. (2002) *J. Immunol.* **169**, 4745–4751
34. Gorgani, N. N., Altin, J. G., and Parish, C. R. (1999) *Int. Immunol.* **11**, 1275–1282
35. Saigo, K., Shatsky, M., Levitt, L. J., and Leung, L. K. (1989) *J. Biol. Chem.* **264**, 8249–8253
36. Rylatt, D. B., Sia, D. Y., Mundy, J. P., and Parish, C. R. (1981) *Eur. J. Biochem.* **119**, 641–646
37. Rickli, E. E., and Otavsky, W. I. (1975) *Eur. J. Biochem.* **59**, 441–447
38. Wiman, B. (1980) *Biochem. J.* **191**, 229–232
39. Moroi, M., and Aoki, N. (1976) *J. Biol. Chem.* **251**, 5956–5965
40. Stathakis, P., Lay, A. J., Fitzgerald, M., Schlieker, C., Matthias, L. J., and Hogg, P. J. (1999) *J. Biol. Chem.* **274**, 8910–8916
41. Borza, D. B., and Morgan, W. T. (1998) *J. Biol. Chem.* **273**, 5493–5499
42. Brown, K. H. (1998) *Am. J. Clin. Nutr.* **68**, 425S–429S
43. Hayes, M. L., and Castellino, F. J. (1979) *J. Biol. Chem.* **254**, 8777–8780
44. Hayes, M. L., and Castellino, J. F. (1979) *J. Biol. Chem.* **254**, 8768–8771
45. Hayes, M. L., and Castellino, F. J. (1979) *J. Biol. Chem.* **254**, 8772–8776
46. Saez, C. T., Jansen, G. J., Smith, A., and Morgan, W. T. (1995) *Biochemistry* **34**, 2496–2503
47. Aktulga, A., and Ulutin, O. N. (1974) in *Recent Advances in Basic Research and Clinical Aspects* (Ulutin, O. N., and Jones, J. V., ed) pp. 185–191, Excerpta Medica, Amsterdam
48. Gorodetsky, R., Mou, X., Blankenfeld, A., and Marx, G. (1993) *Am. J. Hematol.* **42**, 278–283
49. Morgan, W. T. (1978) *Biochim. Biophys. Acta* **535**, 319–333
50. Jones, A. L., Hulett, M. D., and Parish, C. R. (2004) *J. Biol. Chem.* **279**, 30114–30122
51. Borza, D. B., and Morgan, W. T. (1997) *J. Biol. Chem.* **272**, 5718–5726
52. Walther, P. J., Hill, R. L., and McKee, P. A. (1975) *J. Biol. Chem.* **250**, 5926–5933
53. Reich, E. (1978) in *Biological Markers of Neoplasia: Basic and Applied Aspects* (R. W. Ruddon, ed) pp. 491–500, Elsevier, New York
54. Blasi, F. (1999) *Thromb. Haemostasis* **82**, 298–304
55. Sappino, A. P., Belin, D., Huarte, J., Hirschel-Scholz, S., Saurat, J. H., and Vassalli, J. D. (1991) *J. Clin. Invest.* **88**, 1073–1079
56. Scherrer, A., Wohlwend, A., Kruithof, E. K., Vassalli, J. D., and Sappino, A. P. (1999) *Br. J. Haematol.* **105**, 920–927

**Glycobiology and Extracellular Matrices:
Plasminogen Is Tethered with High Affinity
to the Cell Surface by the Plasma Protein,
Histidine-rich Glycoprotein**

GLYCOBIOLOGY AND
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